

Imon Kalyan Ghosh

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📍 Magdeburg, Germany

🌐 LinkedIn

🐙 Github

👤 PROFILE

- **Full-stack AI Engineer** with 3+ years of professional experience in software development, currently pursuing a Master's degree in **Data Science** with a focus on **Generative AI**
- Skilled in **API and LLM integration**, **workflow automation**, **RAG** and **multi-agent workflows**, **deployment** and **monitoring** of AI systems
- Strong communication skills with structured and independent way of working

📁 PERSONAL PROJECTS

AI-Powered SQL & Visual Analytics Platform – (Link)

April 2026

- Developed an end-to-end AI analytics platform that transforms natural language input into optimized SQL queries, actionable insights and interactive data visualizations for any connected **Supabase** environment
- Enabled secure schema extraction and query execution via **MCP** connectivity
- Orchestrated a multi-agent backend for SQL and chart generation, KPI recommendations and metadata-aware reasoning
- Set up **Redis** caching and **SQS**-based async pipelines to offload heavy workloads from latency-sensitive request flows
- Built a scalable serverless **AWS** architecture via **Lambda**, API Gateway, **S3** and **CloudFront** for full-stack deployment
- Automated infrastructure provisioning and release workflows using **Terraform** and **Github Actions CI/CD**

Multimodal RAG Chatbot – (Link)

March 2026

- Built a RAG application (**FastAPI** + **React**) for querying over websites, multiple file types, audio, and embedded images
- Created an ingestion pipeline with **Crawl4AI** for intelligent web-crawling and **Docling** for parsing and chunking
- Designed a structured knowledge storage using **PostgreSQL** for embeddings, metadata, and semantic search
- Implemented a **PydanticAI** RAG agent with tool routing to enable adaptive and hybrid retrieval strategies
- Performed evaluation using **DeepEval** (>90% on all critical metrics), observability using **Langfuse** and **Cloudwatch**
- Containerized via **Docker**, pushed to Amazon **ECR** and deployed to AWS **AppRunner**

Job search Automation pipeline using n8n - (Link)

Jan 2026

- Built a job-search automation workflow using **n8n**, integrating LinkedIn data ingestion, AI-based relevance scoring, and automated resume tailoring
- Utilized Apify and Google API to scrape and log structured job data to a personal Google Sheets repository
- Created a chatbot as a n8n **web-hook** trigger, that captures a LinkedIn job URL and outputs relevant job postings

👛 PROFESSIONAL EXPERIENCE

HiWi SOFTWARE DEVELOPER, Otto von Guericke University

June 2025 – Current | Magdeburg, Germany

- Developed and enhanced full-stack features for the OVGU university app using **Spring Boot/Kotlin**, **TypeScript**, **Flutter**, and **PostgreSQL**, improving user engagement by **20%**
- Performed **Linux**-based server migrations and maintained **containerized** services across multiple environments
- Managed **Android** and **IOS** releases, including packaging, app store submission, and updates
- Handled end-to-end **deployment**, **monitoring**, and incident response to ensure high availability and reliable operation

SOFTWARE ENGINEER, Nomura Research Institute Financial Technologies India Pvt. Ltd.

Aug 2020 – Feb 2024 | Kolkata, India

- Developed **critical features** for a **multi-tenant trade back-office system**, processing over A\$4.7 billion in trades per day on the Australian Stock Exchange
- Developed real-time trading dashboards and **25+ end-to-end modules** using **Java**, **Spring Boot**, **SQL** and **JavaScript**
- Actively participated in **Agile-based** sprint planning and **code reviews**, prepared **impact analysis reports** and maintained **technical documentation** to support development and release activities
- Automated **AWS-based deployment** workflows with Jenkins, cutting manual effort by **50%**
- Fixed high priority bugs and enhanced overall system reliability

ACADEMIC PROJECTS

Decentralized Multi-Agent Swarm Reasoning for Constrained Planning Tasks - [\(Link\)](#) Feb 2026

- Designed and implemented a decentralized multi-agent reasoning framework using **LangGraph**, combining Tree-of-Thought search and Ant Colony Optimization for constrained planning tasks
- Built a task-aware scoring pipeline to validate scheduling constraints, rank candidate reasoning traces, and support iterative pheromone-guided search across agents
- Evaluated the system on the NaturalPlanner benchmark, improving solve rate from 24.75% for a single-agent baseline to 40.00% with the full multi-agent framework, demonstrating effectiveness of decentralized agentic planning

Fine-Tuned Open AI's Whisper ASR Model on Low Resource Language - [\(Link\)](#) April 2025

- Fine-tuned OpenAI Whisper model for low-resource Indian languages, achieving improved transcription quality and 100% WER reduction compared to baseline
- Performed comparative analysis of 3 major **PEFT** techniques, with **QLoRA** consistently delivering the best performance
- Developed a configurable CLI-based fine-tuning pipeline that automates the full ML workflow, from data preparation and training to evaluation and model publishing

EDUCATION

Master of Science In Data and Knowledge Engineering,

Otto-von-Guericke-University Magdeburg, Current GPA: 1.5 (German scale)

Apr 2024 – present

Magdeburg, Germany

Bachelor of Technology in Information Technology,

TechnoIndia Saltlake, CGPA: 8.35/10

Aug 2016 – Jul 2020

Kolkata, India

SKILLS SUMMARY

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| • Programming Languages: | Python, R, SQL, Java, Kotlin, JavaScript, TypeScript, Dart |
| • Deep Learning Frameworks: | PyTorch, TensorFlow |
| • LLM Orchestration: | LangChain, LangGraph, Crew AI, Autogen, PydanticAI, Agents SDK |
| • LLM Tools & Integration: | MCP, A2A, OpenAI API, Hugging Face |
| • AI Coding & Developer Tools: | Codex CLI, Claude Code |
| • Automation Tools: | n8n |
| • Backend, Web & App Development: | Fast API, Spring Boot, React, Vue, Flutter, Streamlit, Gradio |
| • Databases & Vector Stores: | Oracle DB, MySQL, PostgreSQL, pgvector, Chroma, Qdrant |
| • Data Analysis: | NumPy, Pandas, Matplotlib, Plotly, ggplot2 |
| • Observability: | Langfuse, LangSmith, AWS CloudWatch |
| • Caching & Queuing Services : | Redis, Amazon SQS, ActiveMQ |
| • Cloud, DevOps & CI/CD: | Git, Modal, Vercel, Docker, AWS, GCP, Terraform, Github Actions, Jenkins |

LANGUAGES

English (C1) | German (A2) | Bengali (Native)

CERTIFICATES

- LLM Engineering certification from Udemy
- Agentic AI certification from Udemy
- LLMops : Building Production-Ready LLM Systems from Educative